

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – DEBUGGING & OPTIMIZATION TASK

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	Assessment:	Assessment Tool 2 – Debugging & Optimisation Task
Weightage:	35% of CIA	Total Marks:	100
CO2 Statement:	Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.		
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Section / Batch:	CSE	Date:	24/07/2026

Code of Conduct

I _____ H SARAVANA
(192511397) _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student: _____h
saravana_____

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Search Data Structure, Stack	Binary Search, Stack Operations	1 - 5	25

Linked List Data Structure	List Operations	6 – 10	25
Queue Data Structures	Queue Operations	11 - 15	25
Queue Data Structures, Stack Notations	Queue Operations, Expression Evaluation	16 - 20	25
GRAND TOTAL:		100 Marks	

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Debugging Accuracy	20	All errors corrected; code runs flawlessly	Most errors corrected; minor issues remain	Partial correction; code runs with issues	Code fails or major errors persist
Error Identification	30	Accurately identifies all logical, syntax, and edge-case errors	Identifies most major errors	Identifies some errors but misses key issues	Struggles to identify errors
Code Quality & Readability	20	Clean, modular, well documented, follows best practices	Mostly clean and readable	Some structure but inconsistent	Poorly written, hard to understand
Explanation & Justification	20	Clear, logical, and well justified reasoning with complexity analysis	Good explanation with some justification	Limited explanation; lacks depth	No or incorrect explanation
Testing & Validation	10	Comprehensive test cases including edge cases	Adequate test cases	Minimal testing	No testing provided
Faculty In-charge		Course Coordinator		Program Director	
Signature: _____		Signature: _____		Signature: _____	
Date: _____		Date: _____		Date: _____	

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)
QUESTIONS
Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List
5 marks

Q9
Linked List
5 marks

Q10
Queue
5 marks

10/10 answered

Q1 Binary Search

Identify the error in the following snippet.

```
int binarySearch(int arr[], int n, int key) {
    int low = 0, high = n - 1;
    while(low < high) {
        int mid = (low + high) / 2;
        if(arr[mid] == key)
            return mid;
        else if(arr[mid] < key)
            low = mid;
        else
            high = mid;
    }
    return -1;
}
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the

Your Fixed Code

```
1 int binarysearch(int arr[],int n,int key){
2     int low = 0,high = n-1;
3     while(low <= high){
4         int mid = low +(high - low)/2;
5         if(arr[mid]==key){
6             return mid;
7         }else if(arr[mid]<key){
8             low=mid+1;
9         }else{
10            high = mid-1;
11        }
12    }
13    return -1;
14 }
```

Submitted

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1

Binary Search

5 marks



Q2

Stack

5 marks



Q3

Stack

5 marks



Q4

Stack

5 marks



Q5

Stack - Balancing Symbols

5 marks



Q6

Singly Linked List

5 marks



Q7

Stack

5 marks



Q8

Linked List

5 marks



Q9

Linked List

5 marks



Q10

Queue

5 marks



10/10 answered

Q2 Stack

5 marks

Find the error in the stack deletion operation. Justify it.

```
int pop() {  
    if(top == 0) {  
        printf("Underflow");  
        return -1;  
    }  
    return stack[top--];  
}
```

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2 #define MAX 5  
3 int stack[MAX];  
4 int top = -1;  
5 void push(int x) {  
6     if(top == MAX - 1) {  
7         printf("overflow\n");  
8         return;  
9     }  
10    stack[++top] = x;  
11 }  
12 int pop() {  
13     if(top == -1) {  
14         printf("underflow\n");  
15         return -1;  
16     }  
17     return stack[top--];  
18 }  
19 int main() {  
20     push(10);  
21     push(20);  
22     printf("%d\n", pop());  
23     printf("%d\n", pop());  
24     printf("%d\n", pop());  
25     return 0;  
26 }
```

Test Input

stdin input (optional)

Output

```
20  
10  
underflow  
-1
```

Pending manual review

Submitted

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1
Binary Search
5 marks**Q2**
Stack
5 marks**Q3**
Stack
5 marks**Q4**
Stack
5 marks**Q5**
Stack - Balancing Symbols
5 marks**Q6**
Singly Linked List
5 marks**Q7**
Stack
5 marks**Q8**
Linked List
5 marks**Q9**
Linked List
5 marks**Q10**
Queue
5 marks

10/10 answered

Stack

5 marks

Check the snippet and identify the logical error.

```
int stack[5];
int top;
void init() {
    top = 0;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #define MAX 5
3
4 int stack[MAX];
5 int top;
6
7 void init() {
8     top = -1;
9 }
10
11 void push(int x) {
12     if(top == MAX -1) {
13         printf("overflow\n");
14         return;
15     }
16     stack[++top] = x;
17     printf("%d pushed\n", x);
18 }
19
20 int pop() {
21     if(top == -1) {
22         printf("underflow\n");
23         return -1;
24     }
25     return stack[top--];
26 }
27
28 int main() {
29     init();
30     int n, x;
31     scanf("%d", &n);
32     for(int i = 0; i < n; i++) {
33         scanf("%d", &x);
34         push(x);
35     }
36     while(top != -1) {
37         printf("%d popped\n", pop());
38     }
39     return 0;
40 }
```

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Test Input

```
3
10
20
30
```

Output

```
10 pushed
20 pushed
30 pushed
30 popped
20 popped
10 popped
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1
Binary Search
5 marksQ2
Stack
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Stack - Balancing Symbols
5 marksQ6
Singly Linked List
5 marksQ7
Stack
5 marksQ8
Linked List
5 marksQ9
Linked List
5 marksQ10
Queue
5 marks

10/10 answered

Q4 Stack

5 marks

Identify the errors in the following snippet.

```
if(ch == ' '){  
    while(stack[top] != ' '){  
        postfix[j++] = pop();  
    }  
    pop();  
}
```

🔍 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2 #define MAX 5  
3 int stack[MAX];  
4 int top = -1;  
5  
6 void push(int x){  
7     if( top == MAX - 1){  
8         printf("overflow\n");  
9         return;  
10    }  
11    stack[++top] = x;  
12    printf("%d pushed\n",x);  
13  
14 }  
15 int pop(){  
16     if(top == -1){  
17         printf("underflow\n");  
18         return -1;  
19     }  
20     return stack[top--];  
21  
22 }  
23  
24 int main(){  
25     push(10);  
26     push(20);  
27     printf("%d popped\n",pop());  
28     printf("%d popped\n",pop());  
29     printf("%d popped\n",pop());  
30     return 0;  
}
```

Test Input

stdin input (optional)

Output

```
10 pushed  
20 pushed  
20 popped  
10 popped  
underflow  
-1 popped
```

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🟢 Submitted

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1
Binary Search
5 marks**Q2**
Stack
5 marks**Q3**
Stack
5 marks**Q4**
Stack
5 marks**Q5**
Stack - Balancing Symbols
5 marks**Q6**
Singly Linked List
5 marks**Q7**
Stack
5 marks**Q8**
Linked List
5 marks**Q9**
Linked List
5 marks**Q10**
Queue
5 marks

10/10 answered

Q5 Stack - Balancing Symbols

5 marks

In balancing Bracket, you identify the error.

```
if((ch == '[' && stack[top] == '[') ||  
(ch == ']' && stack[top] == '[')) {  
    pop();  
} else {  
    return 0;  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 int main() {  
4     char exp[100], stack[100];  
5     int top = -1;  
6     int flag = 1;  
7  
8     printf("Enter expression: ");  
9     scanf("%99s", exp);  
10  
11     for (int i = 0; exp[i] != '\0'; i++) {  
12         char ch = exp[i];  
13  
14         if (ch == '(' || ch == '[' || ch == '{') {  
15             stack[++top] = ch;  
16         }  
17         else if (ch == ')' || ch == ']' || ch == '}') {  
18             if (top == -1) {  
19                 flag = 0;  
20                 break;  
21             }  
22  
23             if ((ch == ')' && stack[top] == '(') ||  
24                 (ch == ']' && stack[top] == '[') ||  
25                 (ch == '}' && stack[top] == '{')) {  
26                 top--;  
27             } else {  
28                 flag = 0;  
29                 break;  
30             }  
31         }  
32     }  
33  
34     if (top != -1) flag = 0;  
35  
36     if (flag == 1)  
37         printf("Balanced");  
38     else  
39         printf("Not Balanced");  
40  
41     return 0;  
}
```

Test Input

{({})}

Output

Enter expression: Balanced

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1

Binary Search
5 marks

Q2

Stack
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Stack - Balancing Symbols
5 marks

Q6

Singly Linked List
5 marks

Q7

Stack
5 marks

Q8

Linked List
5 marks

Q9

Linked List
5 marks

Q10

Queue
5 marks

10/10 answered

Q6 Singly Linked List

5 marks

What is the issue?

```
temp = head
while temp.next != NULL:
    print(temp.data)
temp = temp.next
```

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct Node {
5     int data;
6     struct Node *next;
7 };
8
9 int main() {
10     int n, i, x;
11     struct Node *head = NULL, *temp = NULL, *newNode, *tail = NULL;
12
13     scanf("%d", &n);
14
15     for (i = 0; i < n; i++) {
16         scanf("%d", &x);
17
18         newNode = (struct Node *)malloc(sizeof(struct Node));
19         newNode->data = x;
20         newNode->next = NULL;
21
22         if (head == NULL) {
23             head = newNode;
24             tail = newNode;
25         } else {
26             tail->next = newNode;
27             tail = newNode;
28         }
29     }
30
31     temp = head;
32     while (temp != NULL) {
33         printf("%d ", temp->data);
34         temp = temp->next;
35     }
36
37     return 0;
38 }
```

Test Input

5 10 20 25 30

Output

10 20 25 30 20

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1
Binary Search
5 marksQ2
Stack
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Stack - Balancing Symbols
5 marksQ6
Singly Linked List
5 marksQ7
Stack
5 marksQ8
Linked List
5 marksQ9
Linked List
5 marksQ10
Queue
5 marks

10/10 answered

Q7 Stack

5 marks

Point out the error.
POP(stack):
if top == 0:
print("Underflow")
else:
return stack[top--]

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 100
4
5 int stack[MAX];
6 int top = -1;
7
8 int main() {
9     int n, i;
10
11     scanf("%d", &n);
12
13     for (i = 0; i < n; i++) {
14         scanf("%d", &stack[i]);
15     }
16
17     top = n - 1;
18
19     if (top == -1) {
20         printf("Underflow");
21     } else {
22         printf("Popped element: %d\n", stack[top]);
23         top--;
24
25         printf("Stack after pop: ");
26         for (i = 0; i <= top; i++) {
27             printf("%d ", stack[i]);
28         }
29     }
30
31     return 0;
32 }
```

Test Input

10 20 30 40

Output

Popped element: 0
Stack after pop: 10 20 30 40

⚠ Pending manual review

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1
Binary Search
5 marksQ2
Stack
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Stack - Balancing Symbols
5 marksQ6
Singly Linked List
5 marksQ7
Stack
5 marksQ8
Linked List
5 marksQ9
Linked List
5 marksQ10
Queue
5 marks

10/10 answered

Q8 Linked List

5 marks

Identify the errors in the following code

```
while current != NULL:
    current.next = prev
    prev = current
    current = current.next
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct Node {
5     int data;
6     struct Node *next;
7 };
8
9 int main() {
10     int n, i, value;
11     struct Node *head = NULL, *temp = NULL, *newNode;
12     struct Node *prev = NULL, *current, *next;
13
14     if (scanf("%d", &n) != 1) return 0;
15
16     for (i = 0; i < n; i++) {
17         if (scanf("%d", &value) != 1) return 0;
18
19         newNode = (struct Node *)malloc(sizeof(struct Node));
20         newNode->data = value;
21         newNode->next = NULL;
22
23         if (head == NULL) {
24             head = newNode;
25             temp = newNode;
26         } else {
27             temp->next = newNode;
28             temp = newNode;
29         }
30     }
31
32     current = head;
33     while (current != NULL) {
34         next = current->next;
35         current->next = prev;
36         prev = current;
37         current = next;
38     }
39
40     head = prev;
41
42     temp = head;
43     while (temp != NULL) {
44         printf("%d ", temp->data);
```

Test Input

5 10 20 30 40 50

Output

50 40 30 20 10

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

- Q1
Binary Search
5 marks
- Q2
Stack
5 marks
- Q3
Stack
5 marks
- Q4
Stack
5 marks
- Q5
Stack - Balancing Symbols
5 marks
- Q6
Singly Linked List
5 marks
- Q7
Stack
5 marks
- Q8
Linked List
5 marks
- Q9
Linked List
5 marks
- Q10
Queue
5 marks

10/10 answered

Q9 Linked List

5 marks

What is the error in the below code?

```
new.next = head  
head = new
```

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2 #include <stdlib.h>  
3  
4 struct Node  
5 {  
6     int data;  
7     struct Node *next;  
8 };  
9  
10 int main()  
11 {  
12     struct Node *head = NULL, *new, *temp;  
13     int n, i, value;  
14  
15     scanf("%d", &n);  
16  
17     for(i = 0; i < n; i++)  
18     {  
19         scanf("%d", &value);  
20  
21         new = (struct Node *)malloc(sizeof(struct Node));  
22         new->data = value;  
23         new->next = head;  
24         head = new;  
25     }  
26  
27     temp = head;  
28  
29     while(temp != NULL)  
30     {  
31         printf("%d ", temp->data);  
32         temp = temp->next;  
33     }  
34  
35     return 0;  
36 }
```

Submitted

Test Input

5 10 20 30 40 50

Output

50 40 30 20 10

Pending manual review

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1
Binary Search
5 marksQ2
Stack
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Stack - Balancing Symbols
5 marksQ6
Singly Linked List
5 marksQ7
Stack
5 marksQ8
Linked List
5 marksQ9
Linked List
5 marksQ10
Queue
5 marks

10/10 answered

Q10 Queue

5 marks

What is missing in the below pseudocode?

```
DEQUEUE(Q):  
if front == -1:  
    print("Underflow")  
else:  
    return Q[front++]
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 #define MAX 100  
4  
5 int Q[MAX];  
6 int front = 0, rear = 4;  
7  
8 int dequeue() {  
9     if (front > rear) {  
10        printf("Underflow");  
11        return -1;  
12    }  
13  
14    int item = Q[front++];  
15  
16    if (front > rear) {  
17        front = 0;  
18        rear = -1;  
19    }  
20  
21    return item;  
22 }  
23  
24 int main() {  
25     Q[0] = 10;  
26     Q[1] = 20;  
27     Q[2] = 30;  
28     Q[3] = 40;  
29     Q[4] = 50;  
30  
31     printf("%d", dequeue());  
32     return 0;  
33 }
```

Test Input

5 10 20 30 40 50

Output

10

🚫 Pending manual review

📌 Submitted

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1
Stack
5 marksQ2
Circular Queue
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Doubly Linked List
5 marksQ6
Queue
5 marksQ7
Queue - Deque
5 marksQ8
Binary Search
5 marksQ9
Circular Queue
5 marksQ10
Stack - Postfix Expression
5 marks

10/10 answered

Q1 Stack

5 marks

Point out the error?
PEEK(stack);
return stack[top+1]

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 5
4
5 typedef struct {
6     int arr[MAX];
7     int top;
8 } Stack;
9
10 int isEmpty(Stack *s) {
11     return (s->top == -1);
12 }
13
14 int PEEK(Stack *s) {
15     if (isEmpty(s)) return -1;
16     return s->arr[s->top];
17 }
18
19 int main() {
20     Stack s;
21     s.top = -1;
22
23     s.arr[++s.top] = 10;
24     s.arr[++s.top] = 20;
25     s.arr[++s.top] = 30;
26
27     printf("%d", PEEK(s));
28     return 0;
29 }
```

Test Input

stdin input (optional)

Output

30

Pending manual review

Submitted

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1
Stack
5 marks**Q2**
Circular Queue
5 marks**Q3**
Stack
5 marks**Q4**
Stack
5 marks**Q5**
Doubly Linked List
5 marks**Q6**
Queue
5 marks**Q7**
Queue - Deque
5 marks**Q8**
Binary Search
5 marks**Q9**
Circular Queue
5 marks**Q10**
Stack - Postfix Expression
5 marks

10/10 answered

Q2 Circular Queue

5 marks

Identify the issue in the following snippet for Circular Queue in Traffic Signal.

```
rear = (rear + 1) % MAX;
queue[rear] = vehicle;
if(rear == front){
    printf("Queue Full");
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 5
4
5 int queue[MAX];
6 int front = -1, rear = -1;
7
8 int isFull() {
9     return (rear + 1) % MAX == front;
10 }
11
12 int isEmpty() {
13     return front == -1;
14 }
15
16 void enqueue(int vehicle) {
17     if (isFull()) {
18         printf("Queue Full\n");
19         return;
20     }
21
22     if (isEmpty()) {
23         front = rear = 0;
24         queue[rear] = vehicle;
25     } else {
26         rear = (rear + 1) % MAX;
27         queue[rear] = vehicle;
28     }
29 }
30
31 void display() {
32     if (isEmpty()) {
33         printf("Queue Empty\n");
34         return;
35     }
36     printf("Queue Elements: ");
37     int i = front;
38     while (1) {
39         printf("%d ", queue[i]);
40         if (i == rear) break;
41         i = (i + 1) % MAX;
42     }
43     printf("\n");
44 }
```

Test Input

10 20 30 40 50

Output

Queue Full
Queue Elements: 10 20 30 40 50

Pending manual review

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1

Stack
5 marks

Q2

Circular Queue
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Doubly Linked List
5 marks

Q6

Queue
5 marks

Q7

Queue - Deque
5 marks

Q8

Binary Search
5 marks

Q9

Circular Queue
5 marks

Q10

Stack - Postfix Expression
5 marks

10/10 answered

Q3 Stack

5 marks

Point out the error.
if top == MAX:
Overflow

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 5
4
5 int top = -1;
6
7 void push(int arr[], int x) {
8     if (top >= MAX - 1) {
9         printf("Overflow\n");
10        return;
11    }
12    arr[++top] = x;
13 }
14
15 int main() {
16     int arr[MAX];
17     int n, x;
18
19     scanf("%d", &n);
20     for (int i = 0; i < n; i++) {
21         scanf("%d", &x);
22         push(arr, x);
23     }
24
25     return 0;
26 }
```

Test Input

6 10 20 30 40

Output

Overflow

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1
Stack
5 marks**Q2**
Circular Queue
5 marks**Q3**
Stack
5 marks**Q4**
Stack
5 marks**Q5**
Doubly Linked List
5 marks**Q6**
Queue
5 marks**Q7**
Queue - Deque
5 marks**Q8**
Binary Search
5 marks**Q9**
Circular Queue
5 marks**Q10**
Stack - Postfix Expression
5 marks

10/10 answered

Q4 Stack

5 marks

Identify the issue in the below pseudocode.

POP():

if top == 0 → Underflow

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 100
4
5 int stack[MAX];
6 int top = -1;
7
8 void push(int value) {
9     if (top == MAX - 1) {
10         printf("Overflow\n");
11         return;
12     }
13     stack[++top] = value;
14 }
15
16 int pop() {
17     if (top == -1) {
18         printf("Underflow\n");
19         return -1;
20     }
21     return stack[top--];
22 }
23
24 int main() {
25     int value;
26
27     while (scanf("%d", &value) == 1) {
28         push(value);
29     }
30
31
32     while (top != -1) {
33         printf("%d ", pop());
34     }
35
36     printf("\n");
37     return 0;
38 }
```

Submitted

Test Input

10 20 30

Output

30 20 10

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QUESTIONS

Q1
Stack
5 marksQ2
Circular Queue
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Doubly Linked List
5 marksQ6
Queue
5 marksQ7
Queue - Deque
5 marksQ8
Binary Search
5 marksQ9
Circular Queue
5 marksQ10
Stack - Postfix Expression
5 marks

10/10 answered

Doubly Linked List

5 marks

Identify the error in this Doubly Linked List node deletion.

```
void deleteNode(Node* del) {  
    if (del->prev != NULL)  
        del->prev->next = del->next;  
    if (del->next != NULL)  
        del->next->prev = del->prev;  
}
```

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2 #include <stdlib.h>  
3  
4 struct Node  
5 {  
6     int data;  
7     struct Node *prev;  
8     struct Node *next;  
9 };  
10  
11 struct Node *head = NULL;  
12  
13 void insert(int value)  
14 {  
15     struct Node *newNode, *temp;  
16  
17     newNode = (struct Node*)malloc(sizeof(struct Node));  
18     newNode->data = value;  
19     newNode->next = NULL;  
20     newNode->prev = NULL;  
21  
22     if(head == NULL)  
23     {  
24         head = newNode;  
25     }  
26     else  
27     {  
28         temp = head;  
29         while(temp->next != NULL)  
30             temp = temp->next;  
31  
32         temp->next = newNode;  
33         newNode->prev = temp;  
34     }  
35 }  
36  
37 void deleteNode(struct Node *del)  
38 {  
39     if(del == NULL)  
40         return;  
41
```

Test Input

10 20 30 40

Output

Original List: 10 20 30 40
After Deletion: 10 30 40

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1
Stack
5 marksQ2
Circular Queue
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Doubly Linked List
5 marksQ6
Queue
5 marksQ7
Queue - Deque
5 marksQ8
Binary Search
5 marksQ9
Circular Queue
5 marksQ10
Stack - Postfix Expression
5 marks

10/10 answered

Q6 Queue

5 marks

Identify the error in the Queue enqueue logic.

```
void enqueue(int data) {  
    if (rear == MAX) return;  
    queue[rear++] = data;  
}
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 #define MAX 5  
4  
5 int queue[MAX];  
6 int front = -1, rear = -1;  
7  
8 void enqueue(int data)  
9 {  
10     /* Circular queue full condition */  
11     if (front != -1 && (rear + 1) % MAX == front) {  
12         printf("Queue Overflow\n");  
13         return;  
14     }  
15  
16     /* First insertion */  
17     if (front == -1)  
18         front = 0;  
19  
20     rear = (rear + 1) % MAX;  
21     queue[rear] = data;  
22 }  
23  
24 void display()  
25 {  
26     if (front == -1) {  
27         printf("Queue Underflow\n");  
28         return;  
29     }  
30  
31     int i = front;  
32     while (1) {  
33         printf("%d ", queue[i]);  
34         if (i == rear) break;  
35         i = (i + 1) % MAX;  
36     }  
37 }  
38  
39 int main()  
40 {  
41     int n, i, x;  
42  
43     printf("Enter number of elements: ");
```

Test Input

3 10 20 30 40

Output

```
Enter number of elements: Enter  
elements:  
Queue Elements:  
10 20 30
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1
Stack
5 marksQ2
Circular Queue
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Doubly Linked List
5 marksQ6
Queue
5 marksQ7
Queue - Deque
5 marksQ8
Binary Search
5 marksQ9
Circular Queue
5 marksQ10
Stack - Postfix Expression
5 marks

10/10 answered

Q7 Queue - Deque

5 marks

Find the error in this Deque insertFront operation.

```
void insertFront(int data) {  
    if (front == 0) front = MAX - 1;  
    else front = front - 1;  
    deque[front] = data;  
}
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 #define MAX 5  
4  
5 int deque[MAX];  
6 int front = -1, rear = -1;  
7  
8 void insertFront(int data)  
9 {  
10     // Overflow conditions for circular deque  
11     if ((front == 0 && rear == MAX - 1) || (front == rear + 1))  
12     {  
13         printf("Deque Overflow\n");  
14         return;  
15     }  
16  
17     if (front == -1) // deque is empty  
18     {  
19         front = rear = 0;  
20     }  
21     else if (front == 0) // wrap around  
22     {  
23         front = MAX - 1;  
24     }  
25     else  
26     {  
27         front--; // move front backward  
28     }  
29  
30     deque[front] = data;  
31 }  
32  
33 void display()  
34 {  
35     int i;  
36  
37     if (front == -1)  
38     {  
39         printf("Deque is Empty\n");  
40         return;  
41     }  
42  
43     i = front;
```

Test Input

10 20 30 40 50 60

Output

```
Enter number of elements: Enter  
elements:  
Deque Overflow  
Deque Overflow  
Deque Overflow  
Deque Overflow  
Deque Overflow
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1
Stack
5 marksQ2
Circular Queue
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Doubly Linked List
5 marksQ6
Queue
5 marksQ7
Queue - Deque
5 marksQ8
Binary Search
5 marksQ9
Circular Queue
5 marksQ10
Stack - Postfix Expression
5 marks

10/10 answered

Q8 Binary Search

5 marks

Identify the bug in this Recursive Binary Search.

```
int binarySearch(int arr[], int l, int r, int x) {  
    int mid = l + (r - l) / 2;  
    if (arr[mid] == x) return mid;  
    if (arr[mid] > x) return binarySearch(arr, l, mid, x);  
    return binarySearch(arr, mid, r, x);  
}
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 int binarySearch(int arr[], int l, int r, int x)  
4 {  
5     if(l > r)  
6         return -1;  
7  
8     int mid = l + (r - l) / 2;  
9  
10    if(arr[mid] == x)  
11        return mid;  
12  
13    if(arr[mid] > x)  
14        return binarySearch(arr, l, mid - 1, x);  
15  
16    return binarySearch(arr, mid + 1, r, x);  
17 }  
18  
19 int main()  
20 {  
21     int arr[100];  
22     int n, i, x, pos;  
23  
24     printf("Enter number of elements: ");  
25     scanf("%d", &n);  
26  
27     printf("Enter sorted elements:\n");  
28     for(i = 0; i < n; i++)  
29         scanf("%d", &arr[i]);  
30  
31     printf("Enter element to search: ");  
32     scanf("%d", &x);  
33  
34     pos = binarySearch(arr, 0, n - 1, x);  
35  
36     if(pos != -1)  
37         printf("Element found at index %d\n", pos);  
38     else  
39         printf("Element not found\n");  
40  
41     return 0;  
}
```

Test Input

```
7  
10 20 30 40 50 60 70  
50
```

Output

```
Enter number of elements: Enter  
sorted elements:  
Enter element to search: Element  
found at index 4
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1
Stack
5 marksQ2
Circular Queue
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Doubly Linked List
5 marksQ6
Queue
5 marksQ7
Queue - Deque
5 marksQ8
Binary Search
5 marksQ9
Circular Queue
5 marksQ10
Stack - Postfix Expression
5 marks

10/10 answered

Q9 Circular Queue

5 marks

Identify the bug in this Circular Queue dequeue logic.

```
int dequeue() {  
    if (front == -1) return -1;  
    int data = queue[front];  
    front = (front + 1) % MAX;  
    return data;  
}
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 #define MAX 5  
4  
5 int queue[MAX];  
6 int front = -1, rear = -1;  
7  
8 void enqueue(int data)  
9 {  
10     if ((rear + 1) % MAX == front)  
11     {  
12         printf("Queue Overflow\n");  
13         return;  
14     }  
15  
16     if (front == -1)  
17         front = rear = 0;  
18     else  
19         rear = (rear + 1) % MAX;  
20  
21     queue[rear] = data;  
22 }  
23  
24 int dequeue()  
25 {  
26     if (front == -1)  
27         return -1;  
28  
29     int data = queue[front];  
30  
31     if (front == rear)  
32         front = rear = -1;  
33     else  
34         front = (front + 1) % MAX;  
35  
36     return data;  
37 }  
38  
39 void display()  
40 {  
41     if (front == -1)  
42     {
```

Test Input

4
10 20 30 40

Output

```
Enter number of elements: Enter  
elements:  
Queue before dequeue:  
10 20 30 40  
Deleted Element: 10  
Queue after dequeue:
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1
Stack
5 marks**Q2**
Circular Queue
5 marks**Q3**
Stack
5 marks**Q4**
Stack
5 marks**Q5**
Doubly Linked List
5 marks**Q6**
Queue
5 marks**Q7**
Queue - Deque
5 marks**Q8**
Binary Search
5 marks**Q9**
Circular Queue
5 marks**Q10**
Stack - Postfix Expression
5 marks

10/10 answered

Q10 Stack - Postfix Expression

5 marks

Find the bug in the evaluation of a Postfix expression.

```
while (expression[i] != '\0') {  
    if (isdigit(expression[i])) push(expression[i] - '0');  
    else {  
        int op1 = pop();  
        int op2 = pop();  
        // perform operation: result = op1 + op2;  
        push(result);  
    }  
}
```

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 int stack[20];  
4 int top = -1;  
5  
6 int main() {  
7     char exp[20];  
8     int i = 0, op1, op2, res;  
9  
10    printf("Enter postfix expression: ");  
11    scanf("%19s", exp);  
12  
13    while (exp[i] != '\0') {  
14        if (exp[i] >= '0' && exp[i] <= '9') {  
15            stack[++top] = exp[i] - '0';  
16        } else {  
17            op2 = stack[top--];  
18            op1 = stack[top--];  
19  
20            if (exp[i] == '+')  
21                res = op1 + op2;  
22            else if (exp[i] == '-')  
23                res = op1 - op2;  
24            else if (exp[i] == '*')  
25                res = op1 * op2;  
26            else if (exp[i] == '/')  
27                res = op1 / op2;  
28  
29            stack[++top] = res;  
30        }  
31        i++;  
32    }  
33  
34    printf("Result = %d", stack[top]);  
35    return 0;  
36 }
```

Test Input

82-

Output

Enter postfix expression: Result =
6

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Submitted